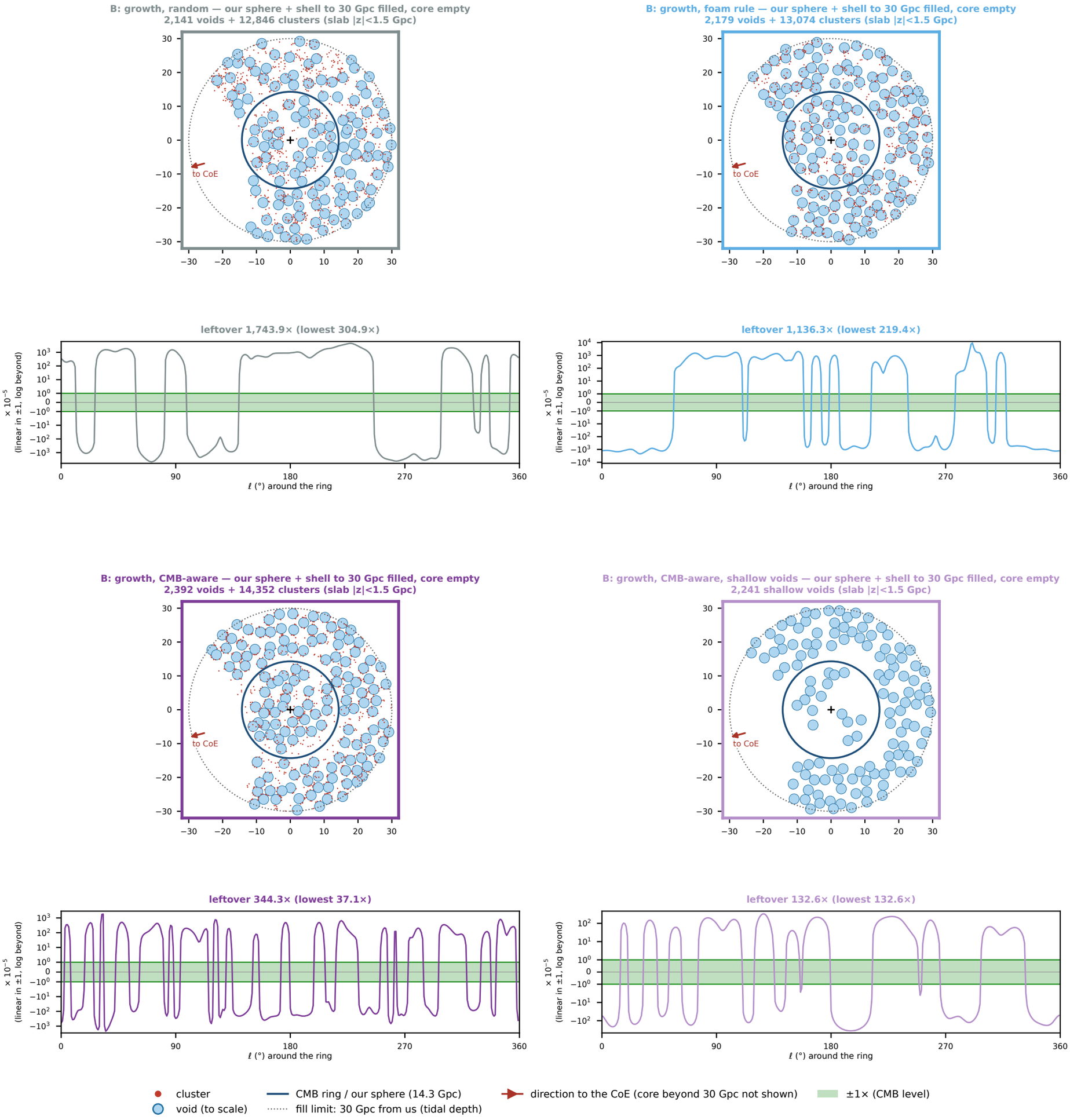

B: growth, random — our sphere + shell to 30 Gpc filled, core empty
2,141 voids + 12,846 clusters (slab |z|<1.5 Gpc)
to CoE
leftover 1,743.9× (lowest 304.9×)
× 10⁻⁵ (linear in ±1, log beyond)
ℓ (°) around the ring
B: growth, foam rule — our sphere + shell to 30 Gpc filled, core empty
2,179 voids + 13,074 clusters (slab |z|<1.5 Gpc)
to CoE
leftover 1,136.3× (lowest 219.4×)
× 10⁻⁵ (linear in ±1, log beyond)
ℓ (°) around the ring
B: growth, CMB-aware — our sphere + shell to 30 Gpc filled, core empty
2,392 voids + 14,352 clusters (slab |z|<1.5 Gpc)
to CoE
leftover 344.3× (lowest 37.1×)
× 10⁻⁵ (linear in ±1, log beyond)
ℓ (°) around the ring
B: growth, CMB-aware, shallow voids — our sphere + shell to 30 Gpc filled, core empty
2,241 shallow voids (slab |z|<1.5 Gpc)
to CoE
leftover 132.6× (lowest 132.6×)
× 10⁻⁵ (linear in ±1, log beyond)
ℓ (°) around the ring
cluster
void (to scale)
CMB ring / our sphere (14.3 Gpc)
fill limit: 30 Gpc from us (tidal depth)
direction to the CoE (core beyond 30 Gpc not shown)
±1× (CMB level)